POUYA HOSSEINI

Home Page — Phone Number — Email — Google Scholar — LinkedIn

Citizenship: Iran
Date of Birth: 26 April 2000

RESEARCH INTERESTS

My research interests lie in the design and optimization of next-generation wireless communication systems (6G and beyond), with a particular focus on:

- Integrated Sensing, Communication, and Computing (ISAC/ISCC) systems with efficient beamforming and resource allocation techniques for joint communication–sensing optimization.
- Internet of Medical Things (IoMT) networks with emphasis on energy efficiency, latency reduction, and reliability in healthcare-oriented wireless systems.
- Semantic-aware for efficient information exchange in resource-constrained environments.
- Application of Artificial Intelligence, including Deep Reinforcement Learning (DRL) and Federated Learning (FL), to enhance the resilience, scalability, and intelligence of next-generation (6G) wireless networks.

EDUCATION

Yazd University *Yazd, Iran*
MSc, Electrical Engineering (Communication Systems) *Sep. 2023 – May 2026*

- GPA: 3.53/4 (16.68/20) (Ranked 1st among 32 students)
- Advisor: Prof. Jamshid Abouei [Home Page] [Google Scholar] [Email]
- Thesis Title: Energy efficiency enhancement in ISCC systems using federated learning and deep reinforcement learning approach
- Thesis Grade: Excellent (20/20)

Yasouj University *Yasouj, Iran*
BSc, Electrical Engineering (Communications) *Sep. 2018 – Jan. 2023*

- GPA: 2.86/4 (14.92/20)
- Advisor: Dr. Sadegh Fadaei [Google Scholar] [Email]
- Thesis Title: Enhancing speed and accuracy of OD-LBP with a novel texture descriptor [Code]

RESEARCH/WORK EXPERIENCE

Yazd University *Yazd, Iran*
Research Assistant *May. 2024 – Present*

- As a Research Assistant in the Wireless Networking Laboratory (WINEL) at Yazd university under supervision of Prof. Jamshid Abouei, I focus on cutting-edge research in wireless communication systems. My work involves exploring advanced technologies such as ISCC, IoMT, FL, Trustworthy AI and energy-efficient wireless networks.

Yazd University *Yazd, Iran*
Teacher Assistant *Sep. 2024 – Jan. 2026*

- Course: Wireless Communications (Under the supervision of Prof. Jamshid Abouei)

Danesh Academy *Ontario, Canada*
Content Creator *Dec. 2023 – Present*

- As an admin, content creator, and editor of the Danesh Academy YouTube channel, We produce educational videos with a primary focus on Electrical Engineering, particularly in the area of wireless communication. [Link]

Sadia Telecommunication Cable Company *Shiraz, Iran*
Engineering Intern *Jun.– Aug. 2022*

- Gained practical experience operating telecommunication switches, such as the NEAX 61, using terminal commands under the guidance of senior engineers.

COURSEWORK

- Principles of Telecommunication Systems, Digital Communication, Discrete-Time Signal Processing, Filter and circuit synthesis, Digital Image Processing, Microwave, Antenna, Telecommunication Circuits, Stochastic Process, Linear Algebra, Communications Networks, Wireless Networks, Information Theory, Digital Video Processing, Wireless Communications, Advanced Communications Theory

TECHNICAL SKILLS

- Programming Languages and Libraries: MATLAB, Python, Pytorch, Tensorflow, C, LaTeX
- Tools: ADS, ANSYS HFSS, Proteus, Symatic Manager (PLC Siemens), NS-3, Cisco Packet Tracer, GNS3, Wireshark
- Others: Machine Learning (ML), Deep Learning (DL), Deep Reinforcemnet Learning (DRL), CompTIA Network+, CCNA, Linux, Content Creation, Microsoft Office, Research Skills, Teamwork

LICENSES & CERTIFICATES

Navid Educational Group *Shiraz, Iran*
English Senior Certificate *Issued Feb. 2017*

Shiraz University *Shiraz, Iran*
Certified MATLAB Professional *Issued Sep. 2019*

Iran Technical & Vocational Training Organization *Shiraz, Iran*
PLC (Grade II) *Issued Aug. 2023*

TOEFL iBT *Yazd, Iran*
Score: 100/120 (5.0/6.0) (R:5.5 L:5.5 W:5.0 S:3.5) *Issued Aug. 2026*

PROJECTS

- Enhancing Low-Quality Facial Images using Super Resolution *Nov. 2024 – Mar. 2025*

LANGUAGES

- English
- Persian

PUBLICATIONS

- **Journal Papers**

1 **P. Hosseini**, S. M. Sheikholeslami, M. Hosseini, J. Abouei, and A. Mohammadi, "Trustworthy federated learning in ISAC-enabled IoMT networks," submitted in *IEEE Internet of Things Journal*, Aug. 2026.

2 **P. Hosseini**, M. M. Kesargheh, J. Abouei, and A. Mohammadi, "SNN-empowered MA-aided ISCC for energy-efficient internet of medical things," submitted in *IEEE Transaction on Wireless Communications*, Jun. 2026.

3 M. Hosseini, **P. Hosseini**, J. Abouei, S. Alikhani, and A. Mohammadi, "Graph-theoretic SIC coordination for RIS-assisted MIMO–NOMA systems using weakly connected dominating sets," submitted in *IEEE Transaction on Wireless Communications*, Jun. 2026.

4 M. Hosseini, **P. Hosseini**, J. Abouei, E. Baladi, and A. Mohammadi, "Energy-efficient resource optimization for UAV-assisted semantic-aware IoMT networks with NOMA," *Elsevier Physical Communications*, May 2026. [Link]

5 M. Mozafari, M. Hosseini, **P. Hosseini**, A. Zahedi, J. Abouei, and A. Mohammadi, "Energy-efficient federated learning for IoT networks with massive MIMO-enabled SWIPT," *IEEE Internet of Things Journal*, Apr. 2026. [Link]

6 M. M. Kesargheh, **P. Hosseini**, N. Nouri, A. Zahedi, J. Abouei, and A. Mohammadi, "Integrated sensing, communication, and computing for energy efficient RIS-aided wireless federated learning," *IEEE Transactions on Vehicular Technology*, vol. 13, no. 5, pp. 9298-9318, Mar. 2026. [Link]

7 G. Kianfar, **P. Hosseini**, M. Azadi, J. Abouei, and A. Mohammadi, "Energy-delay analysis in advection-diffusion-based wireless body area networks", *PLOS ONE*, vol. 20, no. 8, p. 1-20, Aug. 2025. [Link]

- **Conference Papers**

1 S. Fadaei, **P. Hosseini**, K. RahimiZadeh, "New texture descriptor based on improved orthogonal difference local binary pattern", in *Proc. IEEE* 6th *International Conference on Pattern Recognition and Image Analysis (IPRIA)*, Tehran, Iran, Jun, 2023. [Link]

REFERENCES

Prof. Jamshid Abouei
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Faculty of Electrical Engineering, Kermanshah University of Technology, Kermanshah, Iran
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Associate Prof. Elham Baladi
Electrical Engineering Department of Polytechnique Montreal, Montreal, QC, Canada
Email — Personal Home Page — Google Scholar — LinkedIn

Associate Prof. Sadegh Fadaei
Faculty of Electrical Engineering, Yasouj University, Yasouj, Iran
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Dr. Maryam Hosseini
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